

RESEARCH SUMMARY

Topics	Efficient LLM inference	Discrete Diffusion, Flow Maps, Consistency, Inference-Time Scaling, Speculative Decoding
Publications		G-Star — ICML 2026 main track
Skills	Libraries	PyTorch, Hugging Face Transformers (Qwen, LLaMA)
	Training / inference	Multi-node SLURM (DDP, FSDP), DeepSpeed, FlashAttention, bf16, vLLM / SGLang, ONNX
	On-device inference	Pruning, quantization
Background		5+ years in ML (engineer → researcher), 1st year PhD student

PUBLICATIONS

- Guided Star-Shaped Masked Diffusion** — V. Meshchaninov, E. Shibaev, [A. Makoian](#), et al. ICML 2026, main track
- Pretrained masked diffusion LMs fine-tuning for guided correction of already generated tokens.
- Motion ID: Human Authentication Approach** — A. Gavron, et al., [A. Makoian](#) arXiv 2023
- User authentication by motion patterns: $82.2\% \pm 9\%$ accuracy (baseline 70%)

RESEARCH EXPERIENCE

- JetBrains Research** Feb 2025 — Present
ML Researcher / JetBrains Research *with D. Vetrov* Python 3, PyTorch, JAX, transformers, Hydra, Lightning, SLURM
→Masked Diffusion LMs
 - Inference-time scaling, diffusion drafters for speculative decoding; large-scale models (Dream-7B, Qwen3-0.6B MDLM/BD3LM)
→Latent Diffusion Drafters for speculative decoding: lossless acceleration of LLM inference
 - Distillation (flow maps, consistency) and latent space compression for real-time acceleration
- AIRI (Artificial Intelligence Research Institute)** Jul 2022 — Oct 2022
ML Research Intern / GAN adaptation Python 3, PyTorch, Hydra, WandB
 - Few Shot Image Generation implementation for GAN domain adaptation to a new domain from ~10 images
- Samsung R&D Department** Nov 2020 — June 2021
ML Research Intern / Mobile Authentication, see [Motion ID](#) Python 3, PyTorch, Sklearn

EDUCATION

PhD in Computer Science (1st year)	Sep 2026 — Present	Selected courses - Machine Learning - Deep Learning - Generative Models - Bayesian Methods - Optimization Methods - Large-Scale DL - NLP - Computer Vision - Deep Vision 3D Computer Vision - Reinforcement Learning - Probability Theory - Statistics - Linear Algebra - Matrix Computations - Algorithms and DS
Master in Computer Science	Sep 2024 — May 2026	
<i>Constructor University, Bremen, Germany</i>		
Bachelor in Computer Science	Jan 2023 — May 2024	
<i>Neapolis University of Paphos, Cyprus, GPA 86/100</i>		
Bachelor in Computer Science	Sep 2019 — Jun 2024	
<i>HSE University, Moscow, Russia, GPA 8.7/10</i>		
Master's Equivalent in Data Science	Sep 2020 — May 2022	
<i>Yandex School of Data Analysis, Russia</i>		

INDUSTRY EXPERIENCE

- Yandex** Jun 2023 — Sep 2024
Middle ML Engineer/ Geo services Python 3, PyTorch, Hydra, OpenCV, LangChain
 - Map roads segmentation from aerophotos with IoU 0.93; post-processing reduced artifacts from 10% → 0.02%
 - News classification pipeline with LangChain agents for map update hypotheses: precision 0.9, recall 0.85
- Sber Devices** Jul 2022 — Jun 2023
Middle ML Engineer/ Computer Vision Platform Python 3, C++, PyTorch, ONNX, torch_pruning
 - Real-time background removal for video conferencing (0.98 dice); pruning gave 1.5x speed-up
 - On-device 700KB pay card segmentation model with C++ inference in a multiplatform mobile SDK
- Yandex** Jul 2021 — Oct 2021
ML Developer intern / VR Team *with V. Lempitsky* Python 3, PyTorch
 - Trimap-free human alpha matting for VR scenes: improved [FBA Matting](#), built alpha-mask datasets

COMPETITIONS

- Awards in All-Russian Olympiad: 33rd place in **Informatics** and 8th place in **Maths** over 20000 participants Apr 2018, 2019
- Top 8%** in Google Hash Code. **Top 50** in Google Kick Start. **Top 350 (top 0.8%)** in Google Code Jam Feb, May, June 2020